Global Visual Feature and Linguistic State Guided Attention for Remote Sensing Image Captioning

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***Abstract*— The encoder–decoder framework is prevalent in existing remote-sensing image captioning (RSIC) models. The appearance of attention mechanisms brings significant results. However, current attention-based caption models only build up the relationships between the local features without introducing the global visual feature and removing redundant feature components. It will cause caption models to generate descriptive sentences that are weakly related to the scene of images. To solve the problems, this article proposed a global visual feature-guided attention (GVFGA) mechanism. First, GVFGA introduces the global visual feature and fuses them with local visual features to build up their relationships between them. Second, an attention gate utilizing the global visual feature is proposed in GVFGA to filter out redundant feature components in the fused image features and provide more salient image features. In addition, to relieve the hidden state's burden, a linguistic state (LS) is proposed to specifically provide textual features, making the hidden state only guiding visual–textual attention process. What's more, to further refine the fusion of visual features and textual features, a LS-Guided Attention (LSGA) mechanism is proposed. It can also filter out the irrelevant information in the fused visual–textual feature with the help of an attention gate. The experimental results show that this proposed image captioning model can achieve better results on three RSIC datasets, UCM-Captions, Sydney-Captions, and RSICD datasets.**

***Index Terms*— Attention mechanism, image captioning, image description, remote sensing.**

I. Introduction

THE rapid development of launch vehicles and artificial satellite technology makes it easier to observe the earth and lands. At the same time, a large number of remote-sensing images with high spatial resolution are produced. Thus, it is a challenge for people to store, access, and especially find useful information that meets users' needs from so much data. Remote-sensing image captioning (RSIC) is a task spanning from image modality to text modality. It is aimed to understand the content of the input image and generating descriptive sentences to represent the relationships between the objects and the scene of the image. This task can build up the relationships in the image; thus, it has been applied in many essential research fields, such as image understanding and interpretation, image-text retrieval, and intelligent information generation [1]–[5].

Most of the image caption models widely adopt the encoder-decoder framework. The ability of the models to generate sentences is achieved by combining the words predicted by recurrent recursion networks (RNNs) at all decoding time steps, based on the image features extracted by convolutional neural networks (CNNs). Later on, attention mechanism [6] has come out and been applied on encoder–decoder-based models. In the encoder part, self-attention is applied to the encoder part by adjusting the weight of extracted local image features to improve their salience. In the decoder part, soft attention guides the visual–textual attention process to establish the relationships between visual features and textual features. Thus, attention mechanisms have been popular in (NIC) by its advantage of building more relationships within inter- and intra-modality. However, compared with the task of natural image caption, the challenges of remote-sensing image caption are Scale Ambiguity, Category Ambiguity, and Rotation Ambiguity [2].

Many attempts have been made in RSIC tasks; however, there are still some challenges when attention-based methods are applied to RSIC tasks. Firstly, although current attention-based methods [2], [6]–[12], [15], [16] can focus on targets and objects in images, they cannot differentiate features highly or weakly related to the scene of images. The scene of the input image and objects, highly related to the scene, is the main content described in the RSIC task. For example, there are many objects and targets in the airport scene. Among them, the airplane, runway, and terminal are highly related to the airport scene. The grassland, road, and other buildings are not highly related to the airport scene. While the current attention mechanism can focus on many objects and targets, the airplane, runway, and terminal are the targets wanted to be described for the airport scene. Based on this, the global visual feature are introduced to guide the model filter out

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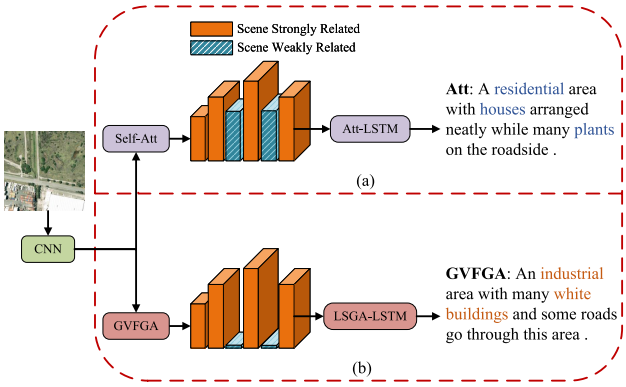


Fig. 1. (a) Self-attention mechanism (Self-Att) and the traditional attention-based LSTM (Att-LSTM). (b) Proposed GVFGA mechanism and the LSGA based LSTM. The GVFGA filters image features weakly related to the image scene and the LSGA refines the visual–textual feature fusion.

features weakly related to the scene of input images, like roads, grassland, and other buildings. What's more, these methods also do not introduce the global visual feature. The lack of global visual feature makes these methods unable to describe the objects' number, shape, and spatial distribution in images from a global and overall perspective. Secondly, these attention-based methods do not consider relieving the burden of hidden state when caption models are predicting words. The hidden state needs to remember the visual information to guide the visual–textual attention process and to remember the textual information to provide textual information for building the context vector. Taking multiple responsibilities will make models' hidden state powerless and increase the error rate when models predict words. Finally, these attention-based methods do not consider further refining the fusion of visual features and textual features. The redundant information stored in the fused visual–textual feature may mislead caption models into predicting wrong words. Fig. 1 shows that the attention on attention (AoA) mechanism [9] introduced at the top can build the relationships between the image features with the help of the self-attention mechanism. It can also filter out the weakly self-related feature components in image features by gating mechanism. But the image features processed by AoA still mix the feature components that are strongly and weakly related to the scene. The weakly related feature component in image features will mislead the caption model into generating words that are less relevant to the scene, thereby reducing the effect of the caption model. In this way, current attention-based models, like AoANet, cannot recognize the weakly related feature components and filter them out.

The remote-sensing images have complex scenes and contain many kinds of objects. Besides, remote-sensing images' descriptions are closely related to scenes of them. For example, if the input image belongs to an airport scene, the description will cover airport-related objects, airplanes, runways, and terminals. Although there may be grassland, meadow, or other objects in the images, these objects are not the primary targets in current remote-sensing images. Describing these objects that are weakly related to the airport will not bring performance improvement to the caption models. Therefore, global visual information is of vital importance in the RSIC task. It can be introduced into the model to build up the relationships between local and global features by fusing them. It can also be used to construct an attention gate to filter out the redundant information in the fused image features [15]. Inspired by this, this article tries to utilize the global visual feature and filter out the redundant information in the image features. Besides, it also tries to filter out the redundant information in the fused visual–textual feature.

Therefore, to alleviate the problems above, this article proposes the global visual feature-guided attention (GVFGA) mechanism and linguistic state-guided attention (LSGA). Firstly, the GVFGA mechanism introduces the global visual feature and fuses them with the local visual features to establish the relationships between them. Moreover, GVFGA also utilizes the global visual feature to construct an attention gate and filters out the feature components that are weakly related to the scene of the image in the image features. Finally, models can generate sentences highly related to the scene of images with the help of fused image features provided by GVFGA. Secondly, a linguistic state (LS) is designed by redesigning the architecture of LSTM, which is specifically designed for providing textual features in each decoding time step. In this way, the burden of the hidden state of LSTM can be relieved. What's more, the hidden state only needs to focus on guiding the visual–textual attention process, avoiding mixing the visual features and textual features. Finally, to further refine the fusion of visual features and textual features, the LSGA mechanism is proposed to guide the visual–textual attention process. LSGA establishes the relationships between visual features and textual features by fusing them. Then, LSGA filters out the redundant information in the fused visual–textual feature, with the help of an attention gate, also constructed by visual and textual features.

A series of experiments are performed on three datasets, UCM-Captions, Sydney-Captions, and RSICD. The experimental results show that the proposed model can achieve better scores on multiple metrics, from which it can be inferred that this method can generate sentences, highly related to remote-sensing images, and avoid predicting wrong words.

In summary, the main contributions in this article are as follows.

1) A GVFGA mechanism is proposed to remove out the redundant information in image features with the help of the attention gate, constructed by the global visual feature.
2) To avoid hidden state mixing visual features up with textual features, a LS is proposed by redesigning the architecture of LSTM to make it specifically providing textual features in each decoding time step.
3) To further refine the fusion of visual features and textual features, a LSGA mechanism is proposed. It utilizes the attention gate to filter out the less essential feature components to construct better the context vector for predicting words.

The rest sections are organized as follows. Section II summarizes the previous work in the field of NIC and RSIC. Section III introduce the proposed GVFGA mechanism and Textual Feature-Guided Attention mechanism. Section IV

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briefly introduce three RSIC datasets, commonly used evaluation metrics, implementation details, and experimental results. Section V conclude the content of this article.

II. Related Works

A. Natural Image Captioning

The rapid development of deep learning in natural scenes brings significant results for the NIC task. At first, Mao *et al*. [7] designed a novel multimodal layer. This layer can connect a CNN subnetwork and an RNN subnetwork to form a whole multimodal recurrent neural network (m-RNN) model, which can extract input image features for setting up the initial state of RNN. Then, with the training process going on, the m-RNN can generate word by word. Finally, by putting all of the words together, a complete sentence can be obtained. It is the firstest attempt at the image captioning task. It achieves the modality conversion, generating sentences for input images with the help of CNN and RNN, and obtain significant performance improvement by being applied to retrieval tasks for retrieving images or sentences. Later, a Neural Image Caption (NIC) model is proposed by Vinyals *et al*. [16]. NIC model can also extract image features by utilizing CNN. Compared with m-RNN, NIC is more adapted to process longer sequences using long-short term memory (LSTM) net [17]. Then, Xu *et al*. [6] designed two novel attention mechanisms, which can be divided into soft attention, and hard attention. In common sense, when human beings express what they see in words, there is a strong correspondence between the area where they focus on the image and the words they express at the moment. Thus, the aim of proposing an attention mechanism is to dynamically generate a weight layer to mimic human beings' ability to focus dynamically on different areas. This operation can lead the model to focus on the corresponding area in the image features according to the input query. Especially when the image features are fed into the decoder, the importance of every pixel location in the image can be adjusted. A bottom-up and top-down framework is proposed by Anderson *et al* [18] for the image captioning task. This method adopts the region-based features, extracted by Faster-RCNN [19], to represent the visual content of the images, which proves that region-based features are more helpful in presenting the relationships between different objects. Later on, AoA is proposed by Huang *et al* [9]. It builds up an information vector and an attention gate to further enhance the relationships within intra-modality. AoA module can refine the image features with the help of a multihead self-attention mechanism and provide more salient features for the decoding process. In recent years, Transformer [20] has been very popular in processing sequential tasks. This method cannot only be competent at sequence tasks, such as, machine translation [21], [22], question answer [23], and text classification [24], but also be applied to many vision tasks, classification [25], object detection [26], [27], and semantic segmentation [28], by regarding image features as sequences and obtain better performance. In the NIC task, the transformer also obtains better performance improvement.

B. Remote-Sensing Image Captioning

There still are several better attempts to generate descriptions for remote-sensing images [3], [29]–[31]. Zhang *et al*. [8] firstly proposed a CNN-RNN based method to generate sentences for describing the content of remote-sensing images. This method utilizes CNN to extract image features from the input images and uses RNN to predict word by word, generating language descriptions. Shi and Zou [32] propose a template-based methods. They use CNN and FCN [33] to interpret the content of images, then compose the interpreted words into a whole sentence with the fixed templates. Lu *et al*. [2] made a significant contribution to the field of RSIC. They published three remote-sensing datasets for the RSIC task and did a series of experiments on these three datasets. They also applied the soft attention mechanism to dynamically adjust the weight of different areas in the images and obtain outstanding performance. An Attribute Attention Model is proposed by Zhang *et al*. [34]. This model can utilize the attribute information extracted from high-level image features to re-weight the image features for providing more salient image features. By introducing the attribute information, the model's ability to generate sentences is improved. Zhang *et al*. [35] firstly designed a novel label-guided attention mechanism, in which method the label information is introduced into the attention calculation process to re-adjust the weight of the attention layer. This operation filters the image information irrelevant to the label of remote-sensing images. More salient image features can be provided to the decoding step, then more accurate sentences can be generated. Later on, Zhang *et al*. [36] proposed another visual aligning attention mechanism. A novel visual aligning loss is designed to improve the salience of the image features in the training phase. Sumbul *et al*. [37] proposed a novel RSIC approach, which is driven by summarization of ground truth captions. This method first adopts the standard CNN-RNN framework to predict words at each time step. Secondly, it utilizes the summarization model to summary five ground truth sentences into one sentence. Finally, it uses the designed adaptive weighting strategy to dynamically attend the predicted word in the standard step and the summarization step. Recently, aiming to alleviate the overfitting problem, Li *et al*. [38] designed a novel truncation cross-entropy loss.

III. Method

In this section, the overview of the proposed framework will be described at first. As shown in Fig. 2, the proposed framework is also based on the encoder-decoder architecture. The encoder can extract image features by fusing the local and global features with the help of the GVFGA module. Moreover, the redundant features can be filtered out. The decoder can feed the refined image features and predict words with the help of the LSGA module. Next, the GVFGA module will be introduced in Part A. After that, the LSGA module will be illustrated in Part B. Finally, the sentences generation process will be described briefly in Part C.

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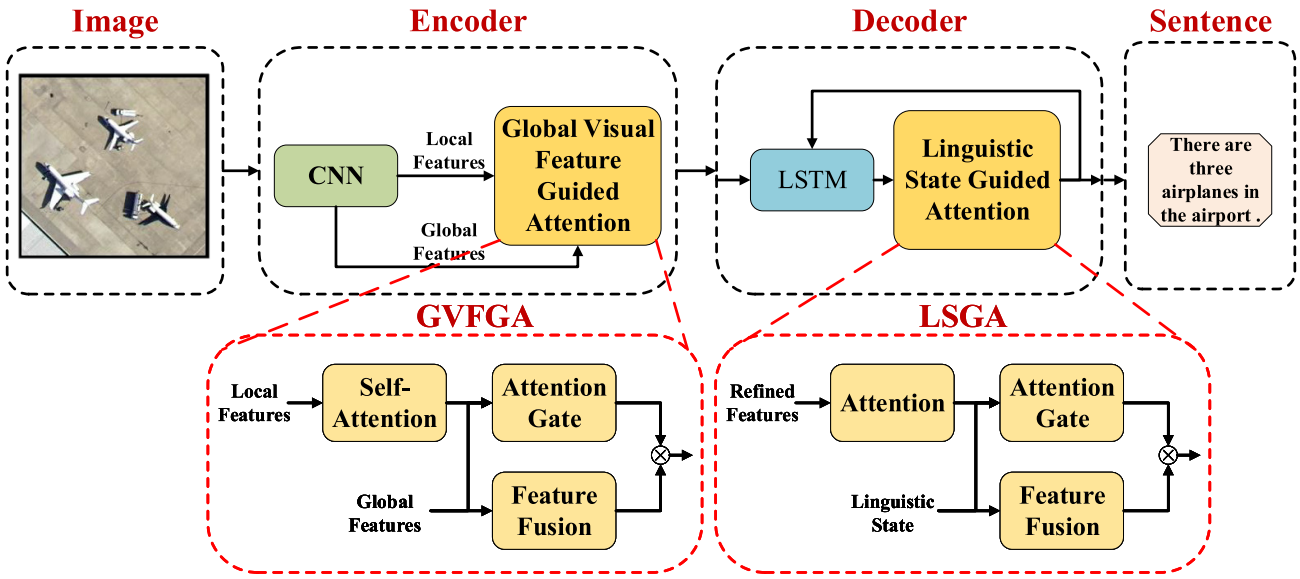


Fig. 2. Framework of the proposed method in this article. In the encoder part, the GVFGA mechanism is designed and adopted. In the decoder part, the LSGA mechanism is designed and adopted.

A. Global Visual Feature-Guided Attention

To further refine the image features, after the self-attention process on the extracted image features, an additional filtering process is needed to be operated on them. Firstly, image features will be extracted from the input images with the help of CNN. Then, a self-attention mechanism will be used on the image features to build up the relationships between the image features. Thus, the multihead self-attention will be briefly introduced firstly. Later on, the proposed GVFGA will be discussed in detail.

1) Multihead Self-Attention: The multihead self-attention mechanism is composed of two modules, multihead, and self-attention. The self-attention mechanism adopts the same similarity calculation method as dot-production to establish the relationships between queries and keys. The difference is that the self-attention additionally adopts a scaling factor $(1/\sqrt{d_k})$. Owing to the similarity calculation is dot-product, the similarity scores of queries and keys are larger if they are similar when the dimension of queries and keys is larger. It will push the region, corresponding to the maximum probability obtained by the softmax function, getting fewer gradients in the backpropagation process [20]. Therefore, this scaling factor is very needed.

The multihead module is to project the d_{model} dimensional queries, keys, and values into h different subspaces by utilizing h trainable linear projections. Then, in each subspace, applying the self-attention mechanism on the Q_i, K_i, and V_i in parallel, yielding d_v dimensional output attended values. These attended values will be concatenated and projected once to obtain the final multihead attention results. The equations are as follows:

$$f_{\text{dot}-\text{att}}(\boldsymbol{Q}_i, \boldsymbol{K}_i, \boldsymbol{V}_i) = \text{softmax}\left(\frac{\boldsymbol{Q}_i \boldsymbol{K}_i^T}{\sqrt{d_k}}\right)\boldsymbol{V}_i \quad (1)$$

$$f_{mh-\text{self}-\text{att}}(\boldsymbol{Q}, \boldsymbol{K}, \boldsymbol{V}) = \text{Concat}(\text{head}_1, \ldots, \text{head}_h) \quad (2)$$

$$\text{head}_i = f_{\text{dot}-\text{att}}\left(\boldsymbol{Q}W_i^Q, \boldsymbol{K}W_i^K, \boldsymbol{V}W_i^V\right) \quad (3)$$

where $\boldsymbol{Q}_i = \boldsymbol{Q}W_i^Q$, $\boldsymbol{K}_i = \boldsymbol{K}W_i^K$, and $\boldsymbol{V}_i = \boldsymbol{V}W_i^V$. $W_i^Q \in \mathbb{R}^{d_{\text{model}} \times d_k}$, $W_i^K \in \mathbb{R}^{d_{\text{model}} \times d_k}$, and $W_i^V \in \mathbb{R}^{d_{\text{model}} \times d_v}$. h denotes the number of heads. $d_k = d_v$ denotes the dimension of subspaces of keys and values. In general $d_k = d_v = d_{\text{model}}/h$. Although the number of subspaces increases, the dimension of each subspace is reduced. Thus, the total computational cost is similar to that of one feature space attention process with d_{model} dimension.

Above all, self-attention can establish the relationships within intra-modality without considering the distance between items in the sequence. Multiheads can improve the ability to represent image features without additional extra computational cost. All the modules mentioned above can make the extracted salient image features.

2) Visual Feature Filtering: Remote-sensing images have complex scenes, cover a large amount of land, and contain many objects. The image features, extracted by normal CNN or further attended by normal attention mechanisms, contain feature components that are strongly and weakly related to the input remote-sensing image scene. Thus, to some extent, those feature components, weakly related to the scene in the image features, are redundant for generating image descriptions. The feature components, strongly related to the scene, are the main features needed to be interpreted and described. Therefore, scene information is of vital importance in the RSIC task. Inspired by this, the GVFGA mechanism is proposed in this article. GVFGA aims to utilize the global visual feature, standing for the input remote-sensing images' scene information, to filter out redundant feature components in the image features. By further refining the image features, GVFGA is able to provide more salient image features for the subsequent decoder process.

Although the self-attention mechanism aims to establish the relationships between image features, it cannot recognize the information weakly related to the scene of remote-sensing

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images and filter it out. Thus, the attended image features needed to be further processed to filter out the redundant information. The proposed GVFGA mechanism considers the characteristics of remote-sensing images. It introduces the global visual feature to filter out redundant information, weakly related to the scene of input images, in image features.

a) Global visual feature extraction: GVFGA mechanism needs to obtain the global visual feature. The most direct way of obtaining the global visual feature is extracting the high-level features from the top network of classification models. However, this way is not very adaptive for processing remote-sensing images. First, the high-level features obtained by fully-connected layers from the top network cannot preserve channel directional information. But the channel directional information is useful, when the global visual features, used to be the query vectors, are multiplied by key vectors. Second, adopting high-level features cannot avoid losing information. For example, for VGG networks, the pixel-level feature's dimension is 512, owing to the number of channels being 512. The dimension of the global visual features needs to be 512, while the dimension of high-level features extracted from the last fully-connected layer is 1000. Thus, dimension reduction is needed to be operated on this 1000-d feature. In this way, even if the high-level features are adopted, the loss of information cannot be avoided. Hence, features obtained by mean pooling are more adaptive than the high-level features from the top network. What's more, the attended image features have been obtained with the help of the attention mechanism. The attended image features belong to pixel-level features, in which each pixel corresponds to one area in the original image. In this article, a mean pooling operation is performed on the image features. It can integrate information of all positions in original images. Therefore, it is regarded as a global visual feature. In addition, the max-pooling is also tried to obtain global visual feature in this article. However, its operation only picks features in one position; thus, it does not integrate information of all locations. This operation cannot obtain good performance and is discarded in this article. The equation is as follows:

$$\boldsymbol{V}_{\text{global}} = f_{\text{mp}}(\boldsymbol{V}_{\text{local}}) \tag{4}$$

where $\boldsymbol{V}_{\text{local}}$ is image features extracted by CNN from the input image. f_{mp} denotes the mean pooling operation.

b) Attention gate and feature fusion: GVFGA mechanism introduces the extra global visual information and utilizes it as the guideline to filter out the redundant information in the image features. To further filter out the redundant information in the image features, the GVFGA mechanism firstly needs to fuse the global visual feature $\boldsymbol{V}_{\text{global}}$ and the attended local features $\hat{\boldsymbol{v}}$ to build up the relationship between them. Later on, the feature components, weakly related to the global image information, can be filtered out by the visual gate $\boldsymbol{g}_{\text{visual}}$, proposed in this article. The visual gate is also composed of attended image features and the global visual feature with the help of the sigmoid function. The flow graph is shown in Fig. 3. It is noteworthy that the local image features $\boldsymbol{V}_{\text{local}}$ need to be processed by the self-attention mechanism $f_{\text{self-att}}$. Then, the attended local image features $\hat{\boldsymbol{v}}$ can be obtained. The

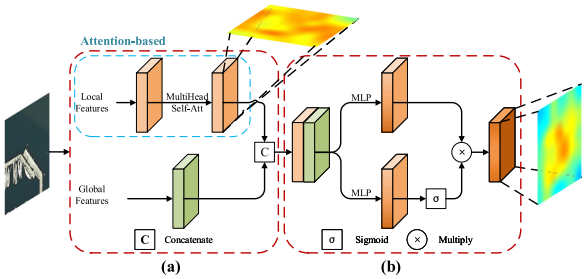


Fig. 3. Whole pipeline of the proposed GVFGA mechanism. Module (a) shows the extraction of local visual features, attended by the multihead self-attention mechanism and the global visual feature. Module (b) presents the feature filtering. The local features and the global features are extracted by CNN from the input images.

specific calculation formulations are as follows:

$$\boldsymbol{V}_{\text{fuse}} = W_g^f \boldsymbol{V}_{\text{global}} + W_v^f \hat{\boldsymbol{v}} + b^f \tag{5}$$

$$\boldsymbol{g}_{\text{visual}} = \sigma\left(W_g^g \boldsymbol{V}_{\text{global}} + W_v^g \hat{\boldsymbol{v}} + b^g\right) \tag{6}$$

where $W_g^f, W_v^f, W_g^g, W_v^g \in \mathbb{R}^{d_{\text{model}} \times d_{\text{model}}}$ are trainable parameters, $b^f, b^g \in \mathbb{R}^{d_{\text{model}}}$ are bias units, and d_{model} is the dimension of $\boldsymbol{V}_{\text{global}}$ and $\hat{\boldsymbol{v}}$, σ denotes the sigmoid activation function.

Here $\hat{\boldsymbol{v}}$ is obtained by self-attention mechanism. The detailed equation is as follows:

$$\hat{\boldsymbol{v}} = f_{\text{self-att}}(\boldsymbol{Q}, \boldsymbol{K}, \boldsymbol{V}) \tag{7}$$

where $f_{\text{self-att}}$ is same as $f_{\text{dot-att}}$ if the attention's process adopting single head mode.

To further obtain the refined image features $\boldsymbol{V}_{\text{refine}}$, the visual gate $\boldsymbol{g}_{\text{visual}}$ is needed to filter out the redundant information in the fused image features $\boldsymbol{V}_{\text{fuse}}$. The equation is as follows:

$$\boldsymbol{V}_{\text{refine}} = \boldsymbol{g}_{\text{visual}} \odot \boldsymbol{V}_{\text{fuse}} \tag{8}$$

where $\odot$ denotes element-wise multiplication. The whole pipeline of the GVFGA mechanism can be formulated as

$$\begin{aligned}\mathcal{GVFGA}(f_{\text{att}}, \boldsymbol{Q}, \boldsymbol{K}, \boldsymbol{V}) \\ = \sigma\left(W_g^g \boldsymbol{q} + W_v^g f_{\text{att}}(\boldsymbol{Q}, \boldsymbol{K}, \boldsymbol{V}) + b^g\right) \\ \odot\left(W_g^f \boldsymbol{q} + W_v^f f_{\text{att}}(\boldsymbol{Q}, \boldsymbol{K}, \boldsymbol{V}) + b^f\right)\end{aligned} \tag{9}$$

where $\boldsymbol{q} = \boldsymbol{V}_{\text{global}}$ and $\boldsymbol{Q} = \boldsymbol{K} = \boldsymbol{V} = \boldsymbol{V}_{\text{local}}$.

B. Linguistic State-Guided Attention

In the attention-based LSTM, the hidden state guides the visual–textual attention process and provides textual features. To relieve the burden of LSTM, an additional textual feature vector is needed to be constructed and be specifically used for providing textual features for the decoding part. Next the proposed LS and LSGA mechanism will be introduced in detail.

1) Construction of Linguistic State: Fig. 4(a) shows the traditional attention-based decoder part. The hidden state needs to guide the visual–textual attention process to make the attention module provide more salient image features. It also needs to provide the textual features for the caption model to ensure that it can predict words based on the visual and

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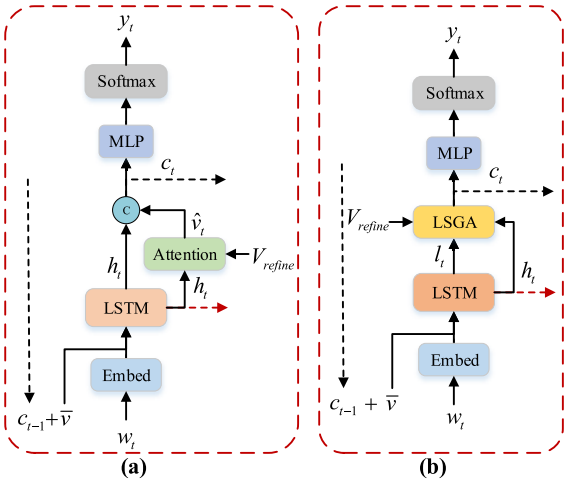


Fig. 4. Module (a) shows the structure of the attention-based LSTM network. Module (b) presents the proposed LSGA mechanism. $\boldsymbol{l}_t$ denotes the LS, storing textual features in the memory of LSTM, designed in this article.

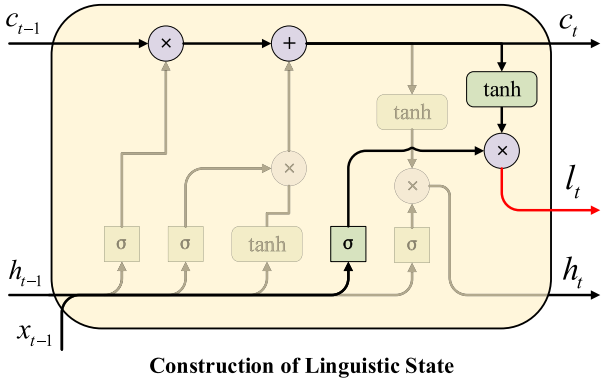


Fig. 5. Structure of LSTM, extended by adding the LS. It shows the construction of the LS. The red line at the right is the output of LS, denoted as $\boldsymbol{l}_t$, at the t time step.

textual information simultaneously. With so many functions at the same time, the hidden state will be powerless. Thus, an additional textual feature vector is needed to be designed to be specifically utilized for providing textual features for the decoding process.

In this article, LSTM is re-designed so that it can provide a new additional vector, the LS $\boldsymbol{l}_t$, as shown in Fig. 5. It can specifically handle the textual features stored in the LSTM and be provided for the subsequent visual–textual feature fusion module. Besides, this LS can relieve the hidden state's burden and improve the LSTM's functions used in the image captioning task.

The LS is constructed modeling on the construction process of the hidden state in the original LSTM. The equations are as follows:

$$\boldsymbol{g}_{l,t} = \sigma\left(W_x^g \boldsymbol{x}_t + W_h^g \boldsymbol{h}_t\right) \tag{10}$$

$$\boldsymbol{l}_t = \boldsymbol{g}_{l,t} \odot \tanh(\boldsymbol{m}_t) \tag{11}$$

where W_x^g, W_h^g are trainable parameters, $\boldsymbol{x}_t$ and $\boldsymbol{h}_t$ denote the input word embedding and hidden states at the t time step, σ denote the sigmoid funciton, $\boldsymbol{m}_t$ denotes the memory of LSTM, $\boldsymbol{l}_t$ denotes textual features extracted from the memory of LSTM, $\odot$ denote element-wise multiplication.

2) Visual and Textual Features Fusion and Filtering: Fig. 4(a) shows that the traditional attention-based decoder part. At the t time step, the context vector is composed of the attended image features and the hidden state of LSTM. These two features is combined by the additive operation. As mentioned above, the hidden state needs to guide the visual–textual attention process so that it cannot specifically focus on providing the textual features stored in the memory of LSTM. Thus, after obtaining the LS $\boldsymbol{l}_t$, specifically used for providing textual features, the textual features and visual features need to be fused to obtain the final context vector. In this article, the LSGA mechanism is adopted to combine the visual and textual features to further improve feature fusion effectiveness. The equations are as follows:

$$\boldsymbol{c}_{\text{fuse},t} = W_l^c \boldsymbol{l}_t + W_v^c \hat{\boldsymbol{v}}_t + b^c \tag{12}$$

$$\boldsymbol{g}_{\text{textual},t} = \sigma\left(W_l^g \boldsymbol{l}_t + W_v^g \hat{\boldsymbol{v}}_t + b^g\right) \tag{13}$$

where W_l^c, W_v^c, W_l^g, W_v^g are trainable parameters, b^c, b^g are bias units, $\boldsymbol{l}_t$, $\boldsymbol{c}_{\text{fuse},t}$, $\boldsymbol{g}_{\text{textual},t}$ are the LS, context vector, textual gate at the t time step, σ is the sigmoid function. $\hat{\boldsymbol{v}}_t$ is the attended results processed by soft-attention mechanism $f_{\text{soft-att}}$. The equation is as follows:

$$\begin{aligned}\hat{\boldsymbol{v}}_t &= f_{\text{soft-att}}(\boldsymbol{Q}, \boldsymbol{K}, \boldsymbol{V}) \\ &= \text{softmax}(W_\alpha \tanh(W_q \boldsymbol{Q} + W_k \boldsymbol{K}))\boldsymbol{V}\end{aligned} \tag{14}$$

where W_α, W_q, W_k are trainable parameters, $\boldsymbol{Q}, \boldsymbol{K}, \boldsymbol{V}$ denote the queries, keys, and values. Here, $\boldsymbol{Q} = \boldsymbol{h}_t$, $\boldsymbol{K} = \boldsymbol{V} = \boldsymbol{V}_{\text{refine}}$.

To obtain the context vector $\boldsymbol{c}_t$, it is necessary to utilize the textual gate $\boldsymbol{g}_{\text{visual}}$ to filter out the redundant information in the $\boldsymbol{c}_{\text{fuse},t}$. The detailed equation is as follows:

$$\boldsymbol{c}_t = \boldsymbol{g}_{\text{textual},t} \odot \boldsymbol{c}_{\text{fuse},t} \tag{15}$$

where $\odot$ represents the element-wise multiplication.

The throughout pipeline of LSGA mechanism can be formulated as

$$\begin{aligned}&\mathcal{LSGA}(f_{\text{soft-att}}, \boldsymbol{Q}, \boldsymbol{K}, \boldsymbol{V}) \\ &\quad= \sigma\left(W_l^g \boldsymbol{q} + W_v^g f_{\text{soft-att}}(\boldsymbol{Q}, \boldsymbol{K}, \boldsymbol{V}) + b^g\right) \\ &\qquad\odot\left(W_l^c \boldsymbol{q} + W_v^c f_{\text{soft-att}}(\boldsymbol{Q}, \boldsymbol{K}, \boldsymbol{V}) + b^c\right)\end{aligned} \tag{16}$$

where $\boldsymbol{q} = \boldsymbol{l}_t$, $\boldsymbol{Q} = \boldsymbol{h}_t$, $\boldsymbol{K} = \boldsymbol{V} = \boldsymbol{V}_{\text{refine}}$.

C. Caption Generation

In this article, the proposed model adopts the encoder-decoder framework [39]. In the encoder part, the proposed GVFGA mechanism is adopted to further filter out redundant information in the attended image features. In the decoder part, an additional LS is proposed to be specifically used for providing textual features. Then, the proposed LSGA mechanism combines the visual features and the textual features. Later on, by combining the predicted words, the final sentences can be obtained.

In this article, the last convolutional layer of the classification model is adopted to extract features from remote-sensing images. It owes to the fact that bounding boxes labels do not

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existed in the existing public RSIC datasets. Thus, R-CNN-based methods [19] are not suitable for extracting object-level features from remote-sensing images. More importantly, there are no independent individual objects in images in some specific remote-sensing scenes, such as forest, ocean, etc. Therefore, in this article, the last convolutional layer of different classification models, such as VGG [40], ResNet [41], are chosen to extract image features, which can be represented by $\boldsymbol{V}_{\text{local}} = \{\boldsymbol{v}_1, \boldsymbol{v}_2, \ldots, \boldsymbol{v}_k\}$

$$\boldsymbol{V}_{\text{local}} = \text{CNN}(\boldsymbol{I}) \tag{17}$$

where $\boldsymbol{I}$ denotes the input images. CNN represents the CNN part of classification models.

As shown in Fig. 4(b), after obtaining the local image features $\boldsymbol{V}_{\text{local}}$, the GVFGA mechanism is applied in the encoder to refine the attended image features further to obtain the refined image features $\boldsymbol{V}_{\text{refine}}$. Then, based on the refine image features, the context vector $\boldsymbol{c}_t$ can be created by the LSGA mechanism, combining the local features and the global visual feature.

LSTM is chosen to be the decoding unit in the decoder part. LSTM's input consists of the word embedding and the visual part at each time step

$$\boldsymbol{x}_t = [W_e w_t, \overline{\boldsymbol{v}} + \boldsymbol{c}_{t-1}] \tag{18}$$

where $\bar{\boldsymbol{v}} = (1/k)\sum_i \boldsymbol{v}_i$ denotes the mean value of $\boldsymbol{V}_{\text{local}}$, obtained by mean pooling operation. $\boldsymbol{c}_{t-1}$ denotes the context vector at the beginning time step, initialized with zeros

$$\boldsymbol{h}_t, \boldsymbol{m}_t = \text{LSTM}(\boldsymbol{x}_t, \boldsymbol{h}_{t-1}, \boldsymbol{m}_{t-1}) \tag{19}$$

where $W_e \in \mathbb{R}^{E\times|D|}$ denotes the word embedding matrix whose size is $|D|$. w_t denotes the word at t time step.

At the t time step, the predicted word's probability distribution $p(\boldsymbol{y}_t)$ can be obtained using linear projection and softmax function. Finally, the predicted word $\boldsymbol{y}_t$ can be obtained by selecting the index corresponding to the largest probability value in the distribution. The equation is as follows:

$$p(\boldsymbol{y}_t) = \text{softmax}(W_p \boldsymbol{h}_t) \tag{20}$$

where $W_p \in \mathbb{R}^{d_{\text{model}}\times|D|}$ is a trainable linear layer.

IV. Experiments

This section will first describe the public RSIC datasets and the common evaluation metrics. After that, details of the implementations will be shown. Finally, the experimental results and analyses will be given.

A. Dataset Description

Three public RSIC datasets, UCM-Captions, Sydney-Captions, and RSICD, are adopted in this article. All the images in these three datasets are shoot by remote satellites and provided for the scene classification task. Later on, Lu *et al.* [2] complemented all these three image datasets with sentences and published them as RSIC datasets. The details of each dataset will be elaborated on below.

1) UCM-Captions: UCM-Captions is a remote-sensing image caption dataset, improved by the UC Merced Land Use Dataset [42], shoot by the National Map of the United States Geological Survey (USGS) [43], and used for scene classification tasks. The number of scene classes is 21, and there are 100 images in each class; the image size is $256 * 256$ pixels. Later on, Lu *et al*. [2] complemented five different descriptive sentences for each image. So, this newly established dataset is called UCM-Captions.

2) Sydney-Captions: Sydney-Captions is also constructed based on the scene classification dataset, the Sydney Dataset [44], in which there are seven classes. The total number of images is 613, and the image size is 500∗500 pixels. In addition, Lu *et al*. [2] also added five different sentence descriptions for each image and created the Sydney-Captions.

3) RSICD: Currently, RSICD is the largest remote-sensing image caption dataset in all these three datasets. It is also proposed by Lu *et al*. [2]. Simultaneously, the images are also shot by satellites and airplanes. There are 10 921 images in RSICD in total, and the image size is $224 * 224$ pixels [3]. There also are five annotations for each image. In the process of creating RSICD, to improve the diversity and accuracy of image annotations, several annotation guidelines are created and followed in interpreting the content of images [2]. Thus, RSICD remote-sensing image caption dataset is accurate in description and complete in content.

4) Datasets Comparison: In Table I, the comparison of all these three remote-sensing image caption datasets is showed. "Average Caption Length" denotes the average length of all sentences in each dataset. "Vocab Size(pre)" and "Vocab Size(post)" denote the size of vocabulary created by each dataset. "Vocab Size(post)" is based on "Vocab Size(pre)" and removes the words that is presented less than five times. Comparing these three datasets shows that RSICD has the most images, the most number of categories, and the most vocabulary size. On the contrary, Sydney-Captions has the least number of images, but its average caption length is the longest, which may bring more difficulties for training models.

B. Evaluation Metrics

Multiple evaluation metrics are adopted in this article to measure the quality of generated image descriptions. In addition, an average evaluation metric is designed to measure the quality of models overall. Next, all of the evaluation metrics will be elaborated on below.

BLEU-n [45] is an evaluation metric, widely used in the machine translation task, used to evaluate the difference between the candidate sentences and reference sentences, ranging from 0.0 to 1.0. BLEU-n is built on the n-gram precision [46], measuring the shared words in the candidate sentences and reference sentences. METEOR [47], [48] is calculated by computing the uni-gram precision and recall probabilities of candidate sentences and reference sentences. It also considers the synonyms and stemmed tokens [49]. ROUGE-L [50] measures the similarity of the candidate sentence and the reference sentence by calculating the longest common subsequence between them [46]. CIDEr [51] is a

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TABLE I
RSIC DATASETS COMPARISON

Datasets	Category Numbers	Average Caption Length	Vocab Size(pre)	Vocab Size(post)	Image Numbers
UCM-Captions	21	11.5	315	298	2100
Sydney-Captions	7	**13.2**	231	179	**613**
RSICD	**30**	11.4	**2695**	**1252**	10000

metric that can measure the candidate sentence and the reference sentence in the vector space. It is achieved by calculating the cosine similarity of Term Frequency Inverse Document Frequency (TF-IDF) vectors [46]. SPICE is a metric that constructs the tuples in the candidate and reference captions and then calculates the F-score of the matched tuples [52]. What's more, SPICE can represents the objects, attributes, and relationships in a graph; thus, it is not sensitive to n-gram.

Since there is not only one evaluation metric for measuring the image caption models, it is needed to proposed an overall metric to evaluate models. Therefore, referring to the equation (21) in AI challenge 2017.[1] An average metric is adopted to measure the quality of models

$$S_m^* = \frac{1}{4}(\text{BLEU}_4 + \text{METEOR} + \text{ROUGE}_L + \text{CIDEr}). \quad (21)$$

Considering the SPICE metric not existing in (21), the overall equation is redesigned by adding it, named S_m. The higher S_m score is, the better image caption model performance is. The detailed definition is as follows:

$$S_m = \frac{1}{5}(\text{BLEU}_4 + \text{METEOR} + \text{ROUGE}_L + \text{CIDEr} + \text{SPICE}) \quad (22)$$

C. Implementation Details

In this article, each dataset is divided into three parts, 80% of the number of image-caption pairs are used for training, 10% of that are used for validating, and 10% of that are used for testing. To avoid that models over-fit or cannot convergent in the training phase, all of the datasets are randomly shuffled. Besides, all of the random seeds are set up at a fixed value for re-productive and fair experiments. In this article, the CNN part has loaded the pretrained models to extract image features of remote-sensing images, trained on ImageNet dataset [53]. In the training phase, the weights of CNN are frozen. Besides, in the multihead self-attention process, the experiments with different h values are explored in Table IV. But, to fairly compare with other methods, the number of heads h is set to 1 in Tables VI–VIII. In addition, for simplicity, h is also set to 1 in Tables II, III, and V. The alpha of Adam optimizer is set to 0.9, and the beta of that is set to 0.999. The learning rate of the encoder is initialized to $5e^{-4}$, and that of the decoder is set to $4e^{-4}$. In addition, in every three epochs, the learning rates will be annealed by a factor of 0.8. All of the experimental codes are based on PyTorch and the models are trained on one 1080Ti GPU.

[1]https://challenger.ai/competition/caption

D. Experimental Results

In this article, three different experiments are designed to prove the effectiveness of the GVFGA module, LSGA module, and the LS. In addition, ablation experiments are performed on UCM-Captions, Sydney-Captions, and RSICD datasets. The detailed experimental results are shown in Table II.

1) Experiments on GVFGA: To prove the effectiveness of GVFGA, several experiments are performed on the above three datasets. The experimental results are shown in Table II. The traditional attention-based models with the AoA module are regarded as the baseline in this article, and the GVFGA module is based on the baseline model. The experimental results show that the GVFGA model can obtain better results than AoA model.

This is due to the fact that the traditional attention-based methods only build up the relationships between the local image features without considering the redundant feature components stored in the image features, which may lead caption models to generate wrong descriptions about input images. Thus, the redundant image features are needed to be removed out. In the encoder part, the GVFGA module introduces the global visual feature into the model, and fuses the global and local visual features. Besides, with the help of the attention gate designed in this article, redundant information stored in the attended image features can be filtered out. Thus, the GVFGA module can further improve the salience of image features, and it can be proved from the experimental results.

In addition, to visually show up the effectiveness of the GVFGA module, heatmaps generated by the GVFGA module and AoA module are shown in Fig. 6. From the heatmaps, it can be seen that GVFGA module filter out the redundant feature components and focus on the areas in the image features that are needed to be described. For example, the original image in the first row in Fig. 6 belongs to the harbor scene, and several ships are in it. Compared with the traditional attention-based methods, the GVFGA module can focus on the central area in the image features. While the traditional attention-based method, the AoA module in the figure, "thinks" that each location in the image features is equally important. The AoA module cannot find the most core location in the image features for the second and third-row images. However, in the image, the GVFGA module can accurately locate the core area, the main object to be described. For the last image in the figure, the AoA module cannot accurately locate the areas to be described. However, the GVFGA module can adaptively focus on the crucial areas in the image features and automatically ignore the bottom-left corner part.

In conclusion, the GVFGA module can remove the redundant image feature components that are weakly related to the

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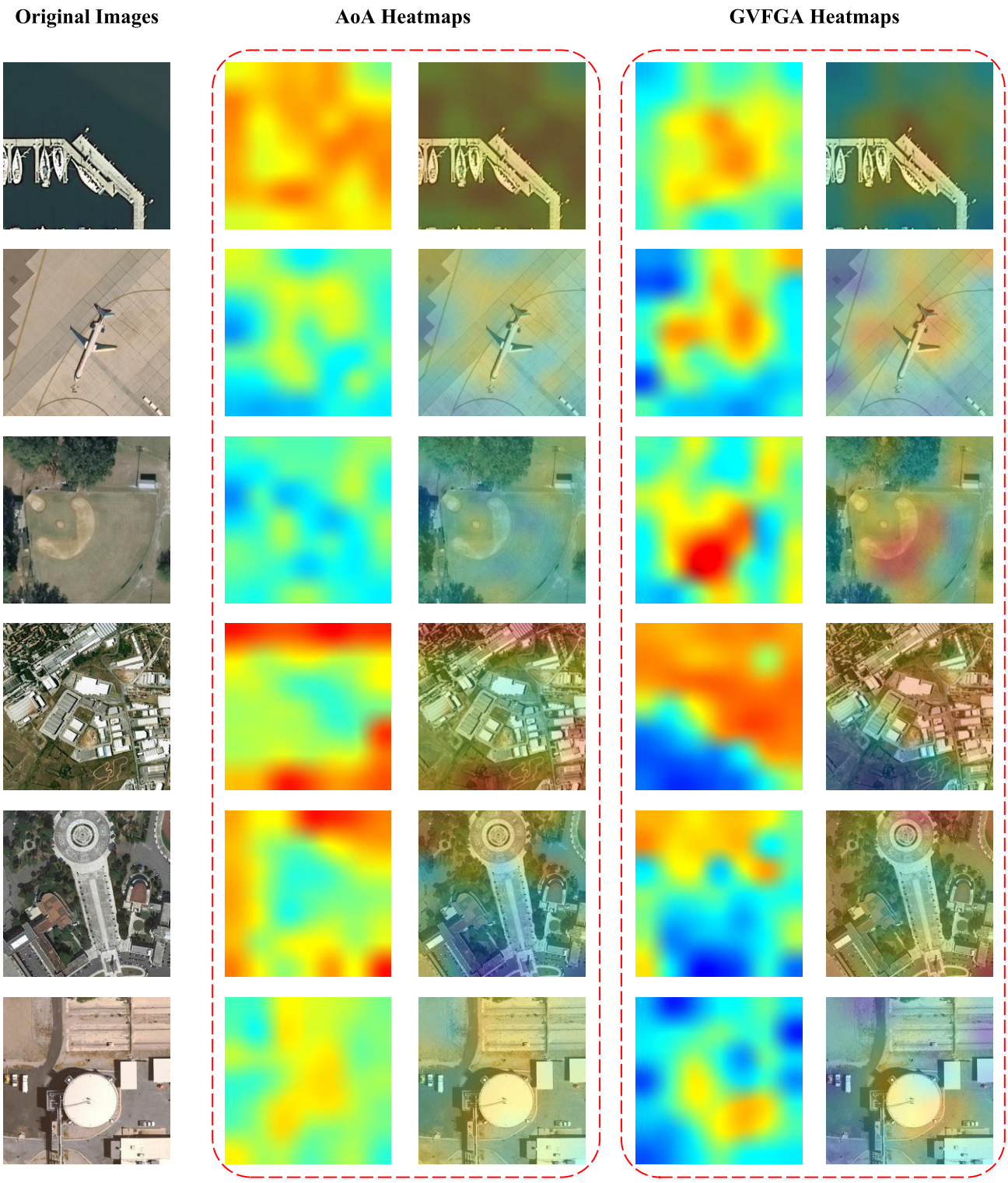


Fig. 6. Comparison of original images, heatmaps generated by AoA module and GVFGA module, and original images added with heatmaps generated by AoA module and GVFGA module.

scene of images and provide more salient image features for the decoder.

2) Experiments on LSGA: To demonstrate the effectiveness of LSGA, several experiments are designed, and the experimental results of models with- and without LSGA module are shown in Table II. The experimental results show that the LSGA module can obtain better results than the AoA model.

In the attention-based LSTM network decoder part, the hidden state of the LSTM network needs to guide the visual–textual attention process. What's more, it is also utilized

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TABLE II
ABLATION EXPERIMENTAL RESULTS ON UCM-CAPTIONS, SYDNEY-CAPTIONS, AND RSICD

Datasets	Ablation GVFGA	Ablation LSGA	B1	B2	B3	B4	M	R	C	S	S_m
UCM-Captions	✓		0.8260	0.7582	0.7012	0.6485	0.4289	0.7741	3.1599	0.4554	1.0934
		✓	0.8165	0.7462	0.6918	0.6423	0.4080	0.7533	3.1260	0.4463	1.0752
	✓	✓	**0.8319**	**0.7657**	**0.7103**	**0.6596**	**0.4436**	**0.7845**	**3.3270**	**0.4853**	**1.1400**
Sydney-Captions	✓		0.7513	0.6633	0.5880	0.5234	**0.3888**	**0.7067**	2.3858	0.4398	0.8889
		✓	0.7517	0.6635	0.5916	0.5300	0.3749	0.6919	2.3617	0.4256	0.8768
	✓	✓	**0.7681**	**0.6846**	**0.6145**	**0.5504**	0.3866	0.7030	**2.4522**	**0.4532**	**0.9091**
RSICD	✓		0.6714	0.5562	0.4760	0.4140	0.3260	**0.5933**	2.5887	**0.4685**	0.8781
		✓	0.6737	0.5543	0.4720	0.4095	0.3247	0.5871	2.5757	0.4661	0.8726
	✓	✓	**0.6779**	**0.5600**	**0.4781**	**0.4165**	**0.3285**	0.5929	**2.6012**	0.4683	**0.8815**

TABLE III
COMPARISON OF HSGA AND LSGA ON UCM-CAPTIONS, SYDNEY-CAPTIONS, AND RSICD

Datasets	Decoder HSGA	Decoder LSGA	B1	B2	B3	B4	M	R	C	S	S_m
UCM-Captions	✓		0.7650	0.6959	0.6409	0.5907	0.3978	0.7299	2.9387	0.4323	1.0179
		✓	**0.8165**	**0.7462**	**0.6918**	**0.6423**	**0.4080**	**0.7533**	**3.1260**	**0.4463**	**1.0752**
Sydney-Captions	✓		0.7361	0.6412	0.5673	0.5050	0.3742	0.6807	2.2309	0.4145	0.8411
		✓	**0.7517**	**0.6635**	**0.5916**	**0.5300**	**0.3749**	**0.6919**	**2.3617**	**0.4256**	**0.8768**
RSICD	✓		0.6662	0.5396	0.4485	0.3797	0.3139	0.5766	2.2711	0.4557	0.7994
		✓	**0.6737**	**0.5543**	**0.4720**	**0.4095**	**0.3247**	**0.5871**	**2.5757**	**0.4661**	**0.8726**

to provide textual features for the visual–textual feature fusion. The hidden state cannot do well in two processed at the same time. The hidden state needs to fetch the visual and textual information from LSTM in guiding the visual–textual process and the visual–textual feature fusion process. Thus, the hidden state will be confused by mixing up with the visual and textual features. However, the LSGA module in the decoder part constructs a novel LS by extending the LSTM network. Moreover, LSGA can guide the visual–textual attention process to obtain fused features. With the help of the gate mechanism, redundant information in the fused features can also be filtered out. The experimental results show that LSGA can further refine the fusion of visual and textual features.

Above all, the LSGA module is effective in guiding the visual–textual attention process. Moreover, the redundant information that is weakly related to each other in establishing the relationships between visual–textual features in the fused features can be filtered out.

3) Experiments on the Linguistic State: To further proving the advantages of LS, additional experiments are performed on UCM-Captions, Sydney-Captions, and RSICD datasets. The experimental results are shown in Table III.

The LSGA module is proposed in this article, and its input is composed of the hidden state and the LS, storing the textual features in the memory of LSTM. Hidden state-guided attention (HSGA) module is designed to prove the effectiveness of LSGA. The difference between the LSGA module and the HSGA module is that HSGA module replaces the LS with the hidden state. In this way, only the hidden state is adopted in the HSGA module.

The experimental results show that the results obtained by the LSGA module are better than those obtained by the HSGA module. Thus, it can be inferred that the hidden state is hard to do well in providing more salient textual features and guiding the visual–textual attention process simultaneously. While the LS indeed plays a role in providing textual features. The LS stores salient textual features and provides more useful textual information to the subsequent visual–textual feature fusion. Simultaneously, introducing the LS relieves the burden of hidden state storing and providing textual features for the visual–textual feature fusion process.

Above all, the LS relief the burden of the hidden state by making it only guiding the visual–textual attention process. What's more, the LS can provide the textual features for the visual-textual features fusion process.

4) Experiments on the Number of Multiheads: Table IV shows that experiments on UCM-Captions, Sydney-Captions, and RSICD datasets with different h values. h denotes the number of multiheads in the multihead self-attention module. It is similar to the number of convolutional kernels. More heads mean more features subspaces with increasing extra parameters and computing costs. In this table, for UCM-Captions and RSICD, more heads can bring better performance improvement. However, for Sydney-Captions, models do not obtain better results with more heads. It may be due to that it contains a small number of image caption pairs.

5) Experiments on the Choice of Global Visual Features: The impact with different choices of global visual features is explored on UCM-Captions, Sydney-Captions, and RSICD datasets, and the results are shown in Table V. GVF denotes

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TABLE IV
EXPERIMENTAL RESULTS WITH DIFFERENT H VALUES

Datasets	h	B1	B2	B3	B4	M	R	C	S	S_m
UCM-Captions	1	0.8319	0.7657	0.7103	0.6596	0.4436	0.7845	3.3270	0.4853	1.1400
	2	0.8253	0.7567	0.6990	0.6469	0.4301	0.7825	3.2130	0.4659	1.1077
	4	**0.8423**	**0.7831**	**0.7339**	**0.6905**	**0.4560**	**0.8000**	**3.4333**	**0.4802**	**1.1720**
	8	0.8413	0.7757	0.7220	0.6732	0.4471	0.7949	3.3756	0.4792	1.1540
Sydney-Captions	1	**0.7681**	**0.6846**	**0.6145**	**0.5504**	**0.3866**	**0.7030**	**2.4522**	**0.4532**	**0.9091**
	2	0.7583	0.6645	0.5886	0.5230	0.3851	0.7066	2.3283	0.4202	0.8726
	4	0.7253	0.6243	0.5459	0.4771	0.3692	0.6640	2.0783	0.4060	0.7989
	8	0.7149	0.6168	0.5433	0.4804	0.3674	0.6621	2.1426	0.4315	0.8168
RSICD	1	0.6779	0.5600	0.4781	0.4165	0.3285	0.5929	2.6012	0.4683	0.8815
	2	0.6790	0.5610	0.4791	0.4155	**0.3312**	**0.5980**	2.6139	**0.4709**	0.8859
	4	0.6627	0.5440	0.4614	0.3995	0.3225	0.5826	2.5397	0.4656	0.8620
	8	**0.6827**	**0.5646**	**0.4833**	**0.4205**	0.3300	0.5961	**2.6437**	0.4708	**0.8922**

TABLE V
EXPERIMENTAL RESULTS WITH DIFFERENT CHOICE OF GLOBAL VISUAL FEATURES

Datasets	GVF	B1	B2	B3	B4	M	R	C	S	S_m
UCM-Captions	MeanPool	0.8319	**0.7657**	**0.7103**	**0.6596**	0.4436	0.7845	**3.3270**	**0.4853**	**1.1400**
	FC	**0.8366**	0.7636	0.7012	0.6438	**0.4439**	**0.7924**	3.2764	0.4760	1.1265
Sydney-Captions	MeanPool	**0.7681**	**0.6846**	**0.6145**	**0.5504**	**0.3866**	**0.7030**	**2.4522**	**0.4532**	**0.9091**
	FC	0.7589	0.6638	0.5895	0.5256	0.3835	0.6983	2.4184	0.4445	0.8941
RSICD	MeanPool	**0.6779**	**0.5600**	**0.4781**	**0.4165**	**0.3285**	**0.5929**	2.6012	**0.4683**	**0.8815**
	FC	0.6741	0.5564	0.4750	0.4130	0.3265	0.5915	**2.6082**	0.4651	0.8809

the choice of obtaining global visual features. $MeanPool$ denotes the global visual features are obtained from the last convolutional layers by meaning pooling operation. FC denotes the global visual features are obtained from the last fully connected layers. In this table, for UCM-Captions, Sydney-Captions, and RSICD, mean pooling operation can obtain better results than high-level features.

E. Comparison With Other Methods

In this article, the proposed method is compared with many other methods in RSIC. The comparison of experimental results are showed in Tables VI–VIII. CSMLF [54] is a retrieval method that recalls the sentences with higher similarities by measuring the distance between the image and collective sentences in the vector space. Xu *et al.* [55] proposed an end-to-end multimodal method that utilizes CNN to extract image feature and regard it as the initial state of the LSTM network. Zhang *et al.* [8] proposed the RNNLM model, utilizing the CaffeNet to obtain multilabels to represent the image scene, then sentences are generated with the help of RNN. Both FC-Att + LSTM and SM-Att + LSTM are based on the attribute attention mechanism [34], which introduces the high-level image features into calculating the attention weights. SAT denotes Show-Attend-and-Tell [6] model, which stands for the traditional attention-based model. SAT feeds the middle-level image features into RNN models at each time step to strengthen the image features' influence on the generation of sentences. GVFGA + LSGA denotes the proposed model adopting the GVFGA mechanism in the encoder part and the LSGA mechanism in the decoding part. It is noteworthy that, in most other methods, the soft attention mechanism is adopted to dynamically adjust image features' weight at different locations. This attention can be as a kind of single head attention. Thus, to make it fair, h is set to 1 in this experimental part. It is wanted to ensure that the improvements are brought by our proposed model instead of being brought by setting more multiheads. Besides, experimental results with multiple evaluation metrics are adopted in Tables VI–VIII to compare the quality of the proposed model and other models. In detail, B1, B2, B3, B4, M, R, C, S represent BLEU-1, BLEU-2, BLEU-3, BLEU-4, METEOR, ROUGE-L, CIDEr, SPICE, separately. S_m is the average metric designed in this article and adopted to measure the quality of generated sentences from the overall perspective.

The experimental results of different methods on UCM-Captions, Sydney-Captions, and RSICD are illustrated in Tables VI–VIII. The methods with $*$ denote the models loaded in the pre-trained weights, fine-tuned on remote-sensing image classification task. Thus, the ability to extract image features of CNN in models with $*$ is higher than the models loaded in the weights of models without $*$ fine-tuning on the ImageNet [56]. It is not hard to find that the GVFGA + LSGA method can get higher scores on S_m metric compared with other methods on the current remote-sensing image caption datasets. The reason for this proposed method being able to obtain the performance improvement on RSIC is that the proposed method considers and utilizes the characteristic of remote-sensing images. The global information of remote-sensing images is of vital importance information in generating descriptive sentences. Although traditional attention-based methods can generate detailed

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TABLE VI
EXPERIMENTAL RESULTS ON UCM-CAPTIONS

Methods	B1	B2	B3	B4	M	R	C	S	S_m
CSMLF[53]	0.4361	0.2728	0.1855	0.1210	0.1320	0.3927	0.2227	-	-
Multimodal[54]	0.7087	0.5966	0.5521	0.4599	0.3462	0.6612	2.9254	-	-
RNNLM*[8]	0.7735	0.7119	0.6623	0.6156	0.4198	0.7233	3.1385	0.4677	1.0730
SAT[6]	0.7995	0.7365	0.6792	0.6244	0.4171	0.7441	3.1044	0.4951	1.0770
FC-Att+LSTM*[33]	0.8102	0.7330	0.6727	0.6188	0.4280	0.7667	3.3700	0.4867	1.1340
SM-Att+LSTM*[33]	0.8115	0.7418	0.6814	0.6296	0.4354	0.7793	3.386	0.4875	1.1436
SAT(LAM)*[34]	0.8195	**0.7764**	**0.7485**	**0.7161**	**0.4837**	**0.7908**	**3.6171**	**0.5024**	**1.2220**
AoANet[9]	0.8185	0.7473	0.6880	0.6327	0.4130	0.7543	3.0873	0.4396	1.0654
GVFGA+LSGA	**0.8319**	0.7657	0.7103	0.6596	0.4436	0.7845	3.3270	0.4853	1.1400

TABLE VII
EXPERIMENTAL RESULTS ON SYDNEY-CAPTIONS

Methods	B1	B2	B3	B4	M	R	C	S	S_m
CSMLF[53]	0.5998	0.4583	0.3869	0.3433	0.2475	0.5018	0.7555	-	-
Multimodal[54]	0.6966	0.6123	0.5431	0.5040	0.3588	0.6347	2.2022	-	-
RNNLM*[8]	0.6861	0.6093	0.5465	0.4917	0.3565	0.6470	2.2129	0.3867	0.8190
SAT[6]	0.7391	0.6402	0.5623	0.5248	0.3493	0.6721	2.2015	0.3945	0.8283
FC-Att+LSTM*[33]	0.7383	0.6440	0.5701	0.5085	0.3638	0.6689	2.2415	0.3951	0.8356
SM-Att+LSTM*[33]	0.7430	0.6535	0.5859	0.5181	0.3641	0.6772	2.3402	0.3976	0.8594
SAT(LAM)*[34]	0.7405	0.6550	0.5904	0.5304	0.3689	0.6814	2.3519	0.4038	0.8673
AoANet[9]	0.7520	0.6620	0.5885	0.5230	0.3792	0.6931	2.2899	0.4209	0.8612
GVFGA+LSGA	**0.7681**	**0.6846**	**0.6145**	**0.5504**	**0.3866**	**0.7030**	**2.4522**	**0.4532**	**0.9091**

TABLE VIII
EXPERIMENTAL RESULTS ON RSICD

Methods	B1	B2	B3	B4	M	R	C	S	S_m
CSMLF[53]	0.5759	0.3859	0.2832	0.2217	0.2128	0.4455	0.5297	-	-
Multimodal[54]	0.6378	0.4756	0.4004	0.3006	0.2905	0.5333	2.2536	-	-
RNNLM*[8]	0.6098	0.5078	0.4367	0.3814	0.2936	0.5456	2.4015	0.4259	0.8096
SAT[6]	0.6707	0.5438	0.4550	0.3870	0.3203	0.5724	2.4686	0.4539	0.8403
FC-Att+LSTM*[33]	0.6671	0.5511	0.4691	0.4059	0.3225	0.5781	2.5763	0.4673	0.8700
SM-Att+LSTM*[33]	0.6699	0.5523	0.4703	0.4068	0.3255	0.5802	2.5738	0.4687	0.8710
SAT(LAM)*[34]	0.6753	0.5537	0.4686	0.4026	0.3254	0.5823	2.5850	0.4636	0.8718
AoANet[9]	0.6718	0.5552	0.4735	0.4101	0.3251	0.5852	2.5647	0.4612	0.8693
GVFGA+LSGA	**0.6779**	**0.5600**	**0.4781**	**0.4165**	**0.3285**	**0.5929**	**2.6012**	**0.4683**	**0.8815**

descriptions with accurate objects and relationships in the images, there still are redundant or wrong descriptions in the generated sentences. Thus, the redundant information in the image features is needed to be filtered out. In this article, the proposed GVFGA module uses the global visual feature as the guideline to guide the attention process in the encoder part to further eliminate redundant information in image features. In the decoder part, the proposed LSGA module is able to fetch more salient textual features from the memory of LSTM and provide them to the subsequent visual–textual feature fusion. Besides, the same attention mechanism is adopted to fuse the visual features and textual features, further filter out the redundant information in the fused features. Above all, the GVFGA module can provide more salient image features without redundant information for the decoder. The LSGA module can provide more textual features to the visual–textual feature fusion process, thereby generating sentences complete in content and rich in semantics.

Different with traditional attention-based models, FC-ATT + LSTM, SM-ATT + LSTM, and LAM methods introduce high-level image features into calculating weights of attention layers. These high-level image feature-guided attention-based methods introduce high-level image information in the extraction of image features and strengthen the relationships between high-level and middle-level image features. Thus, higher scores can be achieved by FC-ATT + LSTM, SM-ATT + LSTM, and LAM methods, compared with SAT model. It is noteworthy that the LAM method implicitly adopts the extra label information to predict more accurate labels of images. More importantly, all these three methods only implement one attention process to adjust the weights of image features. They do not consider trying to filter out redundant information in the image features.

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The AoANet method adopts the AoA mechanism, which implements a second attention process after the first self-attention process to filter out the redundant information, less relevant to the queries in the image features. This method believes that the decoder does not know how the attended image features are related to the query or how much useful information exists in the attended image features. Thus, AoANet designs an information vector, representing the weighted image vectors attended by self-attention, and an attention gate, computed by the sigmoid function. However, it still cannot obtain better performance improvement without considering the global image information of remote-sensing images.

The proposed GVFGA + LSGA model in this article can obtain better performance improvement in RSIC. GVFGA + LSGA model performs attention processes in the encoder and decoder part, respectively. The encoder extracts the global visual feature and introduces them into the features, improving the richness of image features. Besides, it can establish the relationships between the global visual feature and local visual features. This model also filters out the redundant information in the image features by utilizing the gate mechanism. More salient image features can be fed into the decoding part. Simultaneously, in the decoding part, the LSGA module can provide more salient textual features by constructing a LS. Then LSGA can also filter out the redundant information in the fused features, attended by the visual–textual attention process to further improve the correctness of word prediction at each time step. Finally, more accurate and complete sentences can be generated by models.

As we all know, the models trained with a larger dataset can bring better performance improvements. But for image captioning, this rule may not be applicable. From Tables I, VI–VIII, it is not difficult to find that RSICD dataset has the largest quantity of image-caption pairs, but models with the same structure can only obtain the worst performance in all these three datasets. It can be inferred that the vocab size can primarily affect the effectiveness of image caption models. A larger vocab size implies larger searching space, which brings more difficulties for predicting the correct words. Thus, the difficulty brought by the large vocab size is the biggest. RSICD with a larger quantity of dataset only brings a little performance, in contrast to UCM-Captions with almost the same average caption length. What's more, average caption length is also a factor that affects the effectiveness of training models.While Sydney-Captions has the most extended average caption length, it only brings a little performance to models compared to the models trained on the other two datasets.

Above all, it can be seen that GVFGA + LSGA can obtain the best results in the experimental tables. Thus, it can be inferred that the proposed method can effectively provide more salient features, refine the fusion of features, and generate more accurate and complete sentences for remote-sensing images.

F. Comparison of Experimental Results

Fig. 7 shows six representative image description results of two experimental methods in this article. The sentences in the figure are generated by AoANet, regarded as the baseline of RSIC task, and GVFGA + LSGA, proposed in this article. Besides, the "GT" at the bottom represents the reference sentences corresponding to the image. In light of the quality of generated sentences, a detailed analysis will be given below.

Comparing the difference of caption results generated by AoANet and GVFGA + LSGA, it is not hard to find that the sentences generated by GVFGA + LSGA are more accurate and complete in describing global image information, especially on the scene categories, the quantity of objects, and shape of the arrangement shape of areas. Moreover, in contrast to the sentences generated by GVFGA + LSGA, the sentences generated by AoANet are easier to wrongly recognize the category of images and generate the objects not existed in the images.

For the sake of being able to see clearly, the words that can accurately describes the content of images are dyed to red. The wrongly predicted words are dyed to blue. It is easy to see that the first image in the first column is complex in content, containing many colors and objects. The sentence generated by the AoANet method is not accurate, while the GVFGA + LSGA method can generate descriptive sentence representing the content of the image. The objects described by the blue words generated by AoANet do not existed in the image. This result infers that the proposed GVFGA + LSGA method plays a role in eliminating redundant information in image features and visual–textual fused features. For the second image, the GVFGA + LSGA model can accurately describe the number of tennis courts in the image compared with the AoANet method. This result infers that the GVFGA + LSGA method can grasp the overall information of the image so that the number of tennis courts can be predicted more accurately. For the second image in the second column, the GVFGA + LSGA method is also able to describe the shape of the river, "curved," which also proves that the GVFGA + LSGA method can grasp the global image information. For the first image in the second column, the GVFGA + LSGA model can accurately recognize the category of this image, "industrial," and describe it as "white building." In contrast, the AoANet method cannot accurately recognize the category and predict the words. It is because the AoANet method ignores the global image information so that it predicts the "cargo" or "boxes" as "houses." For the last row, the first image is a parking lot with many cars. There is almost no empty parking space. However, for the AoANet method, it generates the sentence with "empty parking space." It is due to that there is redundant information in features provided by the AoANet method. However, the GVFGA + LSGA method can filter out the redundant information in the image features by utilizing the global visual feature. Simultaneously, the LSGA module can also further remove the redundant information in the fused visual–textual features. Thus, the GVFGA + LSGA method can accurately describe the scene of the image overall. For the last image, the AoANet method describes the content of the image wrong. In comparison, the GVFGA + LSGA method can describe the image from the aspect of shape, quantities, and object categories.

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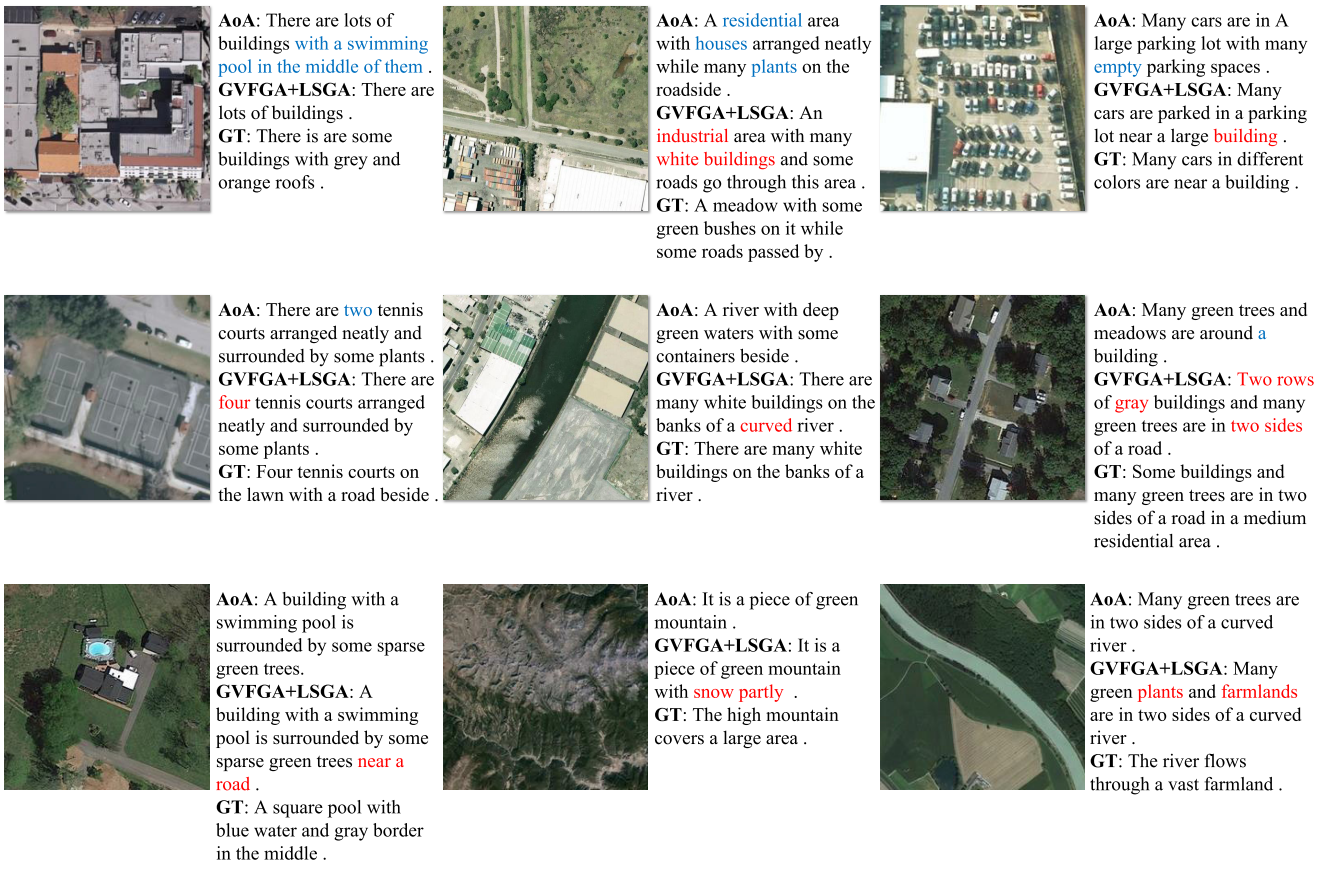


Fig. 7. Each column from left to right is UCM-Captions, Sydney-Captions, and RSICD datasets. On the right side of each image are the sentences generated by AoA and GVFGA + LSGA, and sentences of ground truth, separately. The blue fonts are the wrongly predicted words while the red fonts are the correctly predicted words that can accurately describe the content of images.

In conclusion, the experimental results above show that the GVFGA + LSGA method proposed in this article is effective in RSIC. It can filter out redundant information in the attended image features and visual–textual fused features to improve the quality of features. Finally, better performance improvement can be brought, and higher scores can be achieved. Thus, the GVFGA + LSGA method can generate complete and accurate sentences. Moreover, GVFGA + LSGA can be applied widely in various image captioning models.

V. Conclusion

A GVFGA mechanism and LSGA mechanism are proposed in this article to improve the effectiveness of RSIC models. Firstly, the GVFGA mechanism can fuse the global and local features, both extracted from images, and filter out the redundant information in the fused image features. Secondly, the LSGA mechanism designs a LS containing textual features by extending the LSTM network. It can also fuse the visual and textual features, and filter out the redundant information in the fused visual–textual features. Finally, experimental results show that the GVFGA + LSGA method can obtain further performance improvement. Moreover, this method avoids generating irrelevant descriptive sentences. More importantly, it can also describe the objects' number, shape, and spatial distribution in the image from a global and overall perspective.

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